TRADEWIND TW-1

Solar LoRaWAN field sensor · rev B · 4-layer, 96 × 70 mm

Power

- Solar in: TVS + reverse-FET
- BQ24074 charger, VIN-DPM 4.5 V
- Protected 18650 + NTC
- TPS63900 buck-boost → 3.3 V
- Switched battery sense

File: power.kicad_sch

Radio_MCU

- RAK3172 (STM32WLE5) LoRaWAN
- π-match → U.FL → SMA
- SWD (Tag-Connect), UART pads
- LED, reset, hall-switch wake
- Holes, fiducials

File: mcu_rf.kicad_sch

Sensors_IO

- SHT40 internal T/RH
- J4 air probe (SHT45)
- J3 soil probe, switched 3.3 V
- J2 rain gauge, debounced
- TPD4E02B04 ESD

File: sensors_io.kicad_sch

Power tree: panel 5 V/2 W → VIN_SOLAR → BQ24074 → VSYS (4.4 V in sun, = VBAT in dark) → TPS63900 → +3V3 (module, sensors).

Sleep budget at the battery: BQ24074 4.3 µA + RAK3172 1.7 µA + hall 0.54 µA + rest → about 7 µA.

Default uplink every 5 min (≈ 78 µA average); rapid mode 1 min (≈ 365 µA). Charging allowed 0 to 50 °C (NTC).

All parts on the top side. 50 Ω CPWG RF trace. Fab: 4-layer 1.6 mm (JLC 7628 stackup), ENIG, matte black mask, white silk.

Rev B changes from rev A: high-side battery-sense switch (sleep 34 → 7 µA), debounce C22 22 nF added,

π-match fitted (C14 1.8 pF, L2 7.5 nH) after tuning in the closed case.

Generated from tools/design.py by tools/gen_schematic.py. Checked with KiCad 9 ERC.

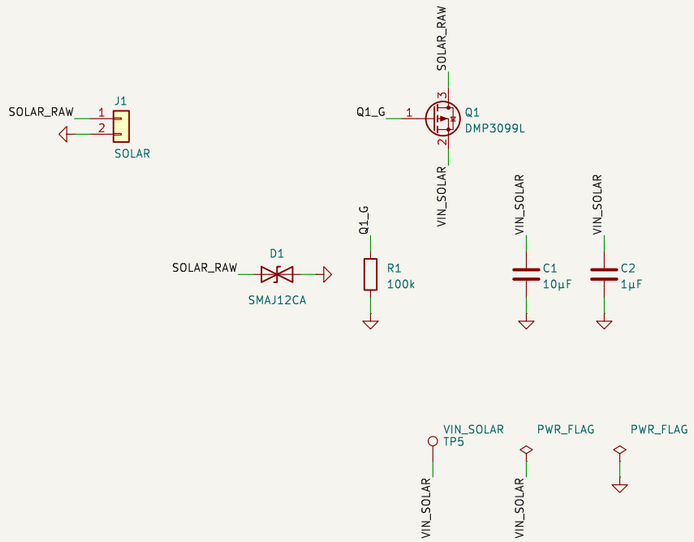
CERN-OHL-P-2.0
Binary Beach reference design
Root sheet
Binary Beach LLC · Honolulu, Hawaii

Sheet: /
File: tradewind.kicad_sch

Title: Tradewind TW-1 · Solar LoRaWAN field sensor

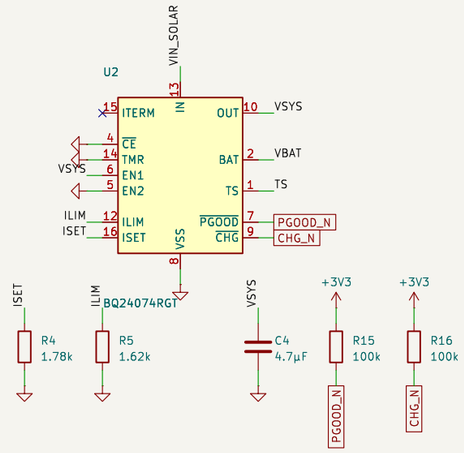
Size: A3	Date: 2026-10-11	**Rev: B**
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SOLAR INPUT & PROTECTION



Panel 5 V / 2 W (Voc 6.0 V). D1 clamps surges, Q1 blocks reverse polarity (Vgs ±20 V).

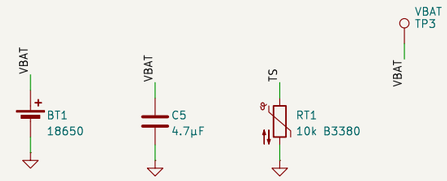
CHARGER + POWER PATH · BQ24074



EN2=L, EN1=H: USB500, 475 mA limit, VIN−DPM 4.5 V (solar−friendly).

TMR=GND: timers off for slow solar charge. RISET 1.78k: 500 mA. ITERM open: 10 %.

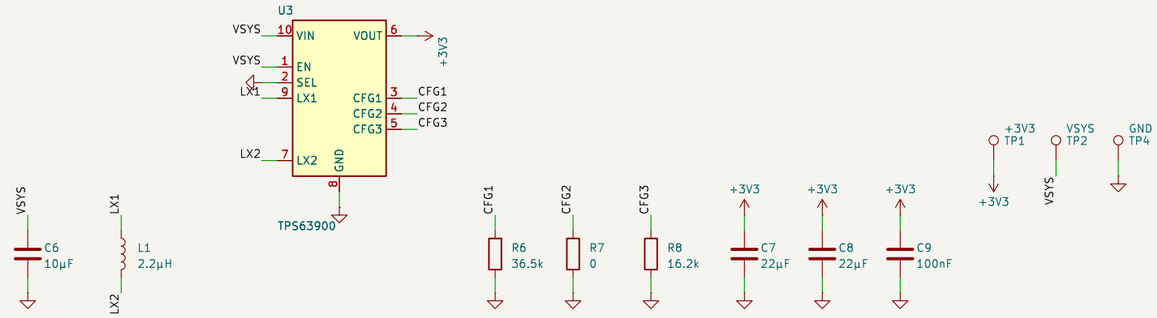
BATTERY · PROTECTED 18650



TS: 10k NTC, ITS 75 µA. Charging only 0 to 50 °C.

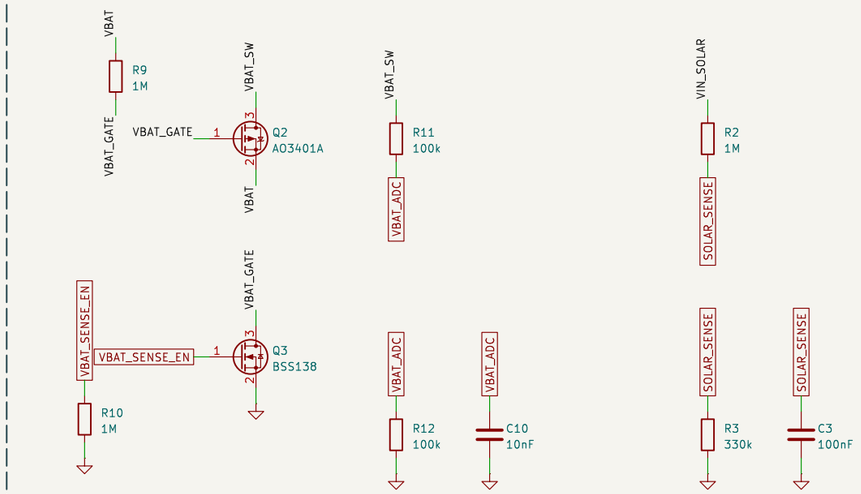
BQ24074 drains 4.3 µA from BAT with no sun (largest sleep load).

3.3 V BUCK−BOOST · TPS63900



RCFG3 16.2k: VO1 = 3.3 V. RCFG1 36.5k + RCFG2 0R: VO2 = 3.3 V, no input limit. SEL = L.

BATTERY & SOLAR SENSE



Rev B: high−side switch. Rev A switched the low side, so VBAT leaked into the ADC clamp (+27 µA at 4.05 V).

Divider 1:2, settle 5 τ = 2.5 ms. Solar divider 1M/330k: ratio 0.248.

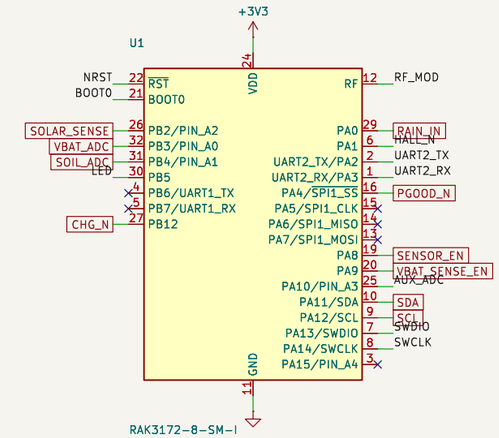
CERN−OHL−P−2.0
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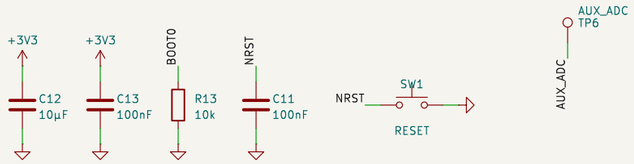
Sheet: /Power/
File: power.kicad_sch

Title: Tradewind TW−1 · Power: solar, charger, battery, 3.3 V

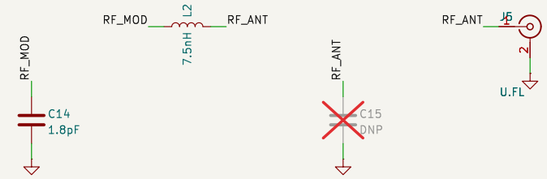
Size: A3	Date: 2026−10−11	**Rev: B**
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LoRaWAN MODULE · RAK3172 (STM32WLE5)





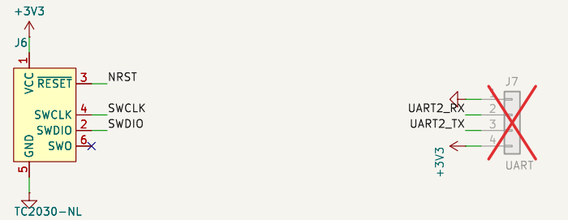
RF MATCH & ANTENNA



50 Ω CPWG on L1 over L2: 0.35 mm trace, 0.20 mm gap, 7628 prepreg 0.21 mm.

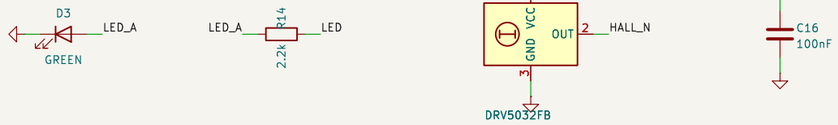
π-match tuned in the closed case: C14 1.8 pF, L2 7.5 nH, C15 DNP (rev A: 0 Ω / DNP).

DEBUG & PROGRAMMING



Flash over UART2 (BOOT0 high) or SWD via Tag-Connect TC2030-NL.

USER INTERFACE · LED + MAGNET WAKE



Tap a magnet on the lid logo: send now. Hold 3 s: rejoin.

MECHANICAL



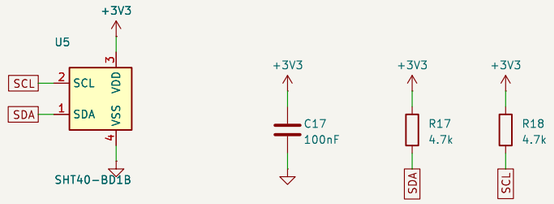
CERN-OHL-P-2.0
Generated from tools/design.py
Solar LoRaWAN field sensor
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Sheet: /Radio_MCU/
File: mcu_rf.kicad_sch

Title: Tradewind TW-1 · LoRaWAN module, RF, debug, UI

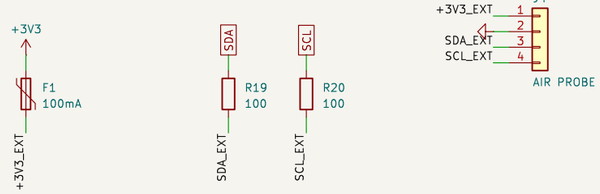
Size: A3	Date: 2026-10-11	**Rev: B**
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INTERNAL T/RH · SHT40 (0x45)



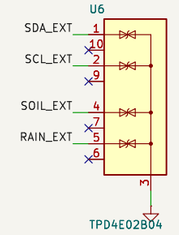
SHT40 sits on a slotted island, away from the charger and regulator.

EXTERNAL AIR PROBE · J4 (SHT45, 0x44)

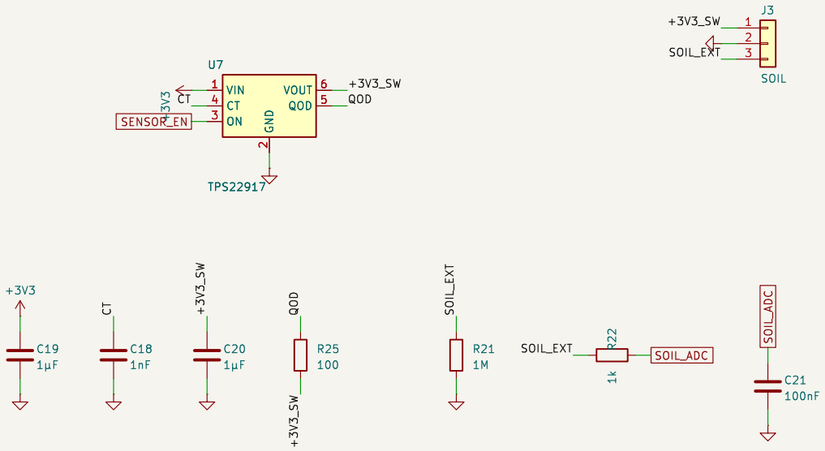


Probe always powered (SHT45 idle 80 nA): a switched probe would clamp the shared I2C bus.

ESD · ALL FIELD LINES

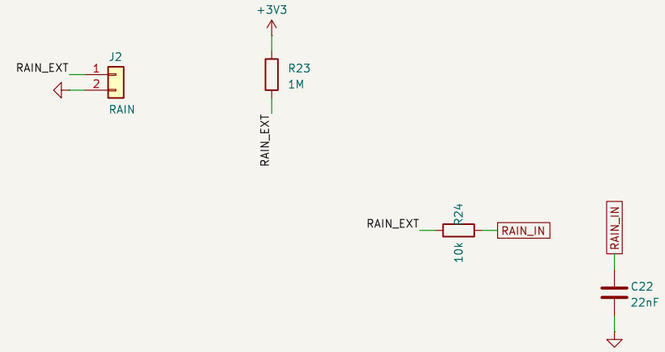


SOIL PROBE · SWITCHED POWER



Probe powered only while sampling: 100 ms warm-up, about 5 mA.

RAIN GAUGE · DEBOUNCED PULSE



Close: τ = 10k·22n = 0.22 ms. Open: τ = 1M·22n = 22 ms. Bounce under 2 ms is absorbed.

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Solar LoRaWAN field sensor
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Sheet: /Sensors_IO/
File: sensors_io.kicad_sch
Title: Tradewind TW-1 · Sensors and field I/O
Size: A3 | Date: 2026-10-11 | **Rev: B**
KiCad E.D.A. 9.0.7 | Id: 4/4